

Model Accuracy Report: Employee Attrition Prediction

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1. Setup

Dataset: IBM HR Analytics (1,470 employees; 16.1% left). Stratified 80/20 split: 1,176 training and 294 test employees (47 leavers in test). Class weights handle the imbalance. Models were compared with 5-fold cross-validation on training data; the test set was used once for the final check.

2. Model comparison (default threshold 0.5, test set)

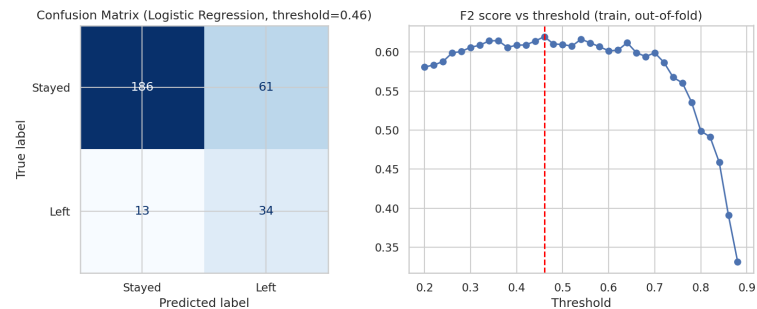
Model	CV ROC-AUC	Accuracy	Precision	Recall	F1	Test ROC-AUC
Logistic Regression (selected)	0.831	0.748	0.345	0.638	0.448	0.806
XGBoost	0.808	0.806	0.407	0.468	0.436	0.772
Random Forest	0.802	0.820	0.312	0.106	0.159	0.778
Decision Tree	0.697	0.765	0.347	0.532	0.420	0.723

Random Forest has the highest accuracy (0.820) but catches only 11% of leavers, which shows why accuracy is misleading on imbalanced data. Logistic Regression was selected for the best cross-validated ROC-AUC (0.831), chosen without looking at the test set.

3. Final model after threshold tuning

The decision threshold was tuned on out-of-fold training predictions by maximising the F2 score (recall counted twice as important as precision), because missing a real leaver costs more than a false alarm. Chosen threshold: **0.46**.

Metric	Accuracy	Precision	Recall	F1	ROC-AUC
Test set (294 employees)	0.748	0.358	0.723	0.479	0.806



	Pred. Stayed	Pred. Left
Actual Stayed	186 (TN)	61 (FP)
Actual Left	13 (FN)	34 (TP)

Confusion matrix: of 47 real leavers the model caught 34 (recall 72%) and missed 13. It raised 61 false alarms among 247 stayers, which is acceptable because a stay-interview costs far less than losing an employee.

4. Interpretation and validation

• **Recall 72%:** about 7 in 10 leavers are flagged in advance. • **Precision 36%:** about 1 in 3 flagged employees really leaves, versus a base rate of 16%, so flagged employees are roughly twice as likely to leave. • **ROC-AUC 0.81:** good separation between leavers and stayers. • **Risk-bucket check (cross-validated, all 1,470 employees):** actual attrition was 37.7% in the High bucket, 11.5% in Medium and 4.2% in Low. • **Limits:** test set has only 47 leavers, so metrics carry uncertainty; dataset is fictional; results show association, not causation.